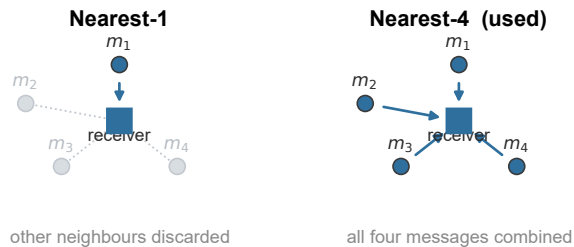


(a)

Message aggregation

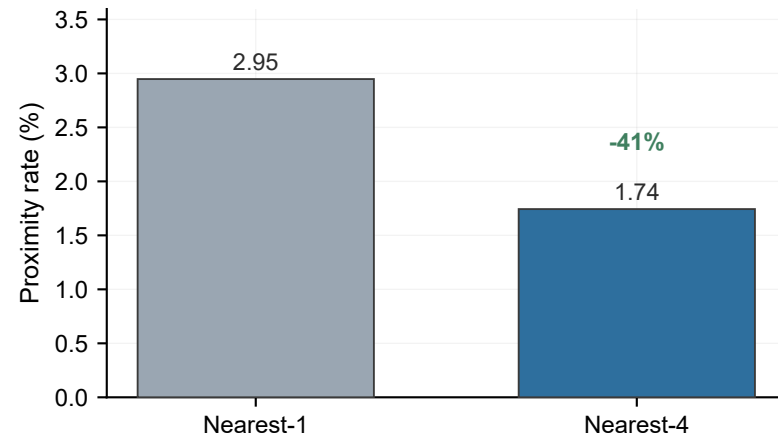


$$[\sin \phi_{ij}, \cos \phi_{ij}, d_{ij}/420] \parallel \mathbf{m}_j \rightarrow f_{\text{loc}} \rightarrow \mathbf{e}_{ij}$$

$$\hat{\mathbf{m}}_i = \frac{1}{|P_i|} \sum_{j \in P_i} \mathbf{e}_{ij} \quad (\text{masked mean})$$

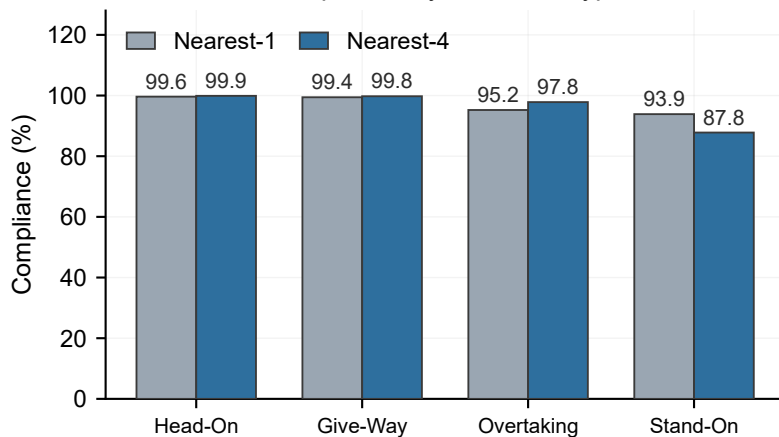
(b)

Proximity rate



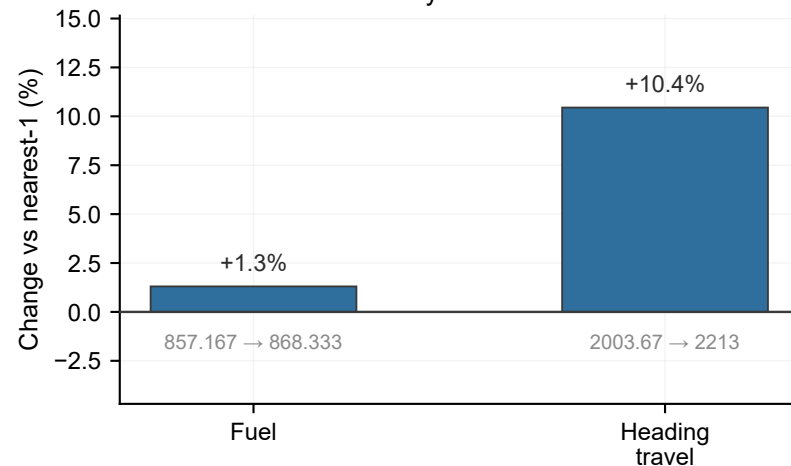
(c)

Compliance by encounter type



(d)

Efficiency vs nearest-1



Arrival 92.0% → 95.3%. PRIMARY v2 compliance 96.5% → 94.6%.
 Nearest-4 is residual+L2 hub; nearest-1 is v11 freeze (not retrained). Direction is the same: four neighbours beat one.